

MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing, Making Connections, and Normal Distributions

Oct 26 – 28 | 4.4 A Closer Look at Testing • 4.5 Making Connections • 5.1 Hypothesis Testing Using Normal Distributions | Fall 2026 • Section 019 • MW 7:30–8:50 PM • RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe Type I and Type II errors in the context of a study, relate α to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CLO 4, CLO 8) 2. Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CLO 8)	e-Text: 4.4 A Closer Look at Testing Interactive Lecture: 4.4 A Closer Look at Testing Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM) Online Homework: 4.4 A Closer Look at Testing (b) (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
3. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping), and compare bootstrap and randomization distributions by their center and use. (CLO 5, CLO 8) 4. Use a 95% confidence interval to reach a two-tailed test conclusion at $\alpha = 0.05$, and decide whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)	e-Text: 4.5 Making Connections Interactive Lecture: 4.5 Making Connections Slides: 4.5 Making Connections	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples	Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
5. Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution $N(\text{null value}, SE)$. (CLO 5) 6. Compute a standardized z statistic and find the p-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CLO 3, CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
<i>Items that cover objectives from more than one row above (objectives in parentheses)</i>			Take-Home Quiz 6 (due Mon Nov 2, 11:59 PM) (all objectives)	Assignment Tool in Blackboard
<i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i>			Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.